

Hermes Architecture

Hermes Agent Architecture

Public-safe architecture snapshot generated at 2026-06-25T06:15:47-04:00.

Source of truth: local Hermes configuration and runtime status on the operator Linux host.

Sensitive values, private chat IDs, OAuth headers, personal finance/client workflow names, and tokens are intentionally omitted or grouped.

Executive Summary

Hermes Agent runs as a **Linux-primary, multi-platform AI operations hub**. The gateway is a systemd service that connects Telegram, Mattermost, WhatsApp, API Server, Home Assistant, MCP servers, cron jobs, skills, memory/context stores, and remote Windows operations.

The default model remains **openai-codex / gpt-5.5**. Local/experimental providers such as **LM Studio** and **Free Kimi** are configured as optional providers only; they are not the default route for Telegram or production workflows.

High-Level Architecture

[!Hermes Architecture](#)

Runtime Topology

Layer	Components	Notes
User channels	Telegram, Mattermost, WhatsApp, API Server	Gateway adapters route inbound messages into the same Hermes agent core.
Agent core	Hermes Agent gateway + CLI + tool loop	Runs on Linux under <code>hermes-gateway.service</code> .
Models	OpenAI Codex primary, Copilot fallback, optional LM Studio, optional Free Kimi proxy	Provider selection is explicit; experimental endpoints are not default.
Tools	Terminal, files, browser, web, vision, TTS, cron, delegation, Home Assistant, GBrain, NotebookLM, CodeGraph	Enabled toolsets differ by platform/session but the gateway exposes the main operational surface.
Knowledge	GBrain, NotebookLM, session search, imported Claude context, skills	Long-term memory and research context are split across structured stores and markdown skills.
Automation	Hermes cron scheduler, no-agent scripts, hooks, plugins	Backups, monitoring, GitHub publication, finance snapshots, smart-home logging, news/media digests, and watchdogs.
Operational surfaces	Tasks, skills, hooks, plugins, MCP servers, toolsets	Documented as public-safe counts/categories plus selected non-sensitive examples.
Remote systems	Windows workstation over SSH, Home Assistant, Google Drive/GitHub, local LM Studio	Linux remains the control plane; Windows is operated remotely when needed.

Operational Surface Inventory

Surface	Detected public-safe state	Notes
Scheduled tasks / cron	36 jobs; 26 no-agent script jobs; 0 agent-backed jobs	Exact private task names are grouped by category.
Skills	226 detected skill files across 22 categories	Private/client-sensitive skill names are omitted from examples.
Hooks / webhooks	shell allowlist present: False; allowlist entries: 0; plugin hook manifests: 23	Hook command bodies are not published.
Plugins	72 visible plugin rows captured; enabled estimate 3	Descriptions omitted to avoid leaking credential/env surfaces.
MCP servers	7 configured MCP servers	GBrain, NotebookLM, CodeGraph are the active core MCP surfaces.

Scheduled tasks / cron categories

Category	Active jobs	Public-safe purpose
Backup & sync	5	Protect configuration, repositories, databases, and knowledge stores.
GitHub & publishing	6	Maintain GitHub/publication surfaces and repo health digests.
Home automation	2	Log smart-home/home-environment telemetry.
Knowledge & memory	4	Keep GBrain/memory/context stores healthy and up to date.
Media/news monitoring	3	News, RSS, YouTube, and briefing pipelines.
Other scheduled automation	5	Other local automation jobs.
Private finance automation	4	Private finance workflow snapshots; details omitted from public docs.
Reliability watchdogs	7	Auto-healing, environment guards, timeout/watchdog checks.

Skills surface

Hermes currently has a broad skill surface. The public inventory lists category counts and non-sensitive examples only.

Skill category	Count
.archive	11
apple	5
autonomous-ai-agents	7
creative	34
data-science	2
devops	8
ecc-imports	4
email	2
gaming	2
github	11
mcp	1
media	7
mlops	19
note-taking	4
personal	6
productivity	23
red-teaming	1
research	15
smart-home	5
social-media	2
software-development	37
uncategorized	20

Public-safe skill examples:

Skill	Category	Public-safe description
dspy	mlops	DSPy: declarative LM programs, auto-optimize prompts, RAG.
fine-tuning-with-trl	mlops	TRL: SFT, DPO, PPO, GRPO, reward modeling for LLM RLHF.
axolotl	mlops	Axolotl: YAML LLM fine-tuning (LoRA, DPO, GRPO).
unsloth	mlops	Unsloth: 2-5x faster LoRA/QLoRA fine-tuning, less VRAM.
huggingface-hub	mlops	HuggingFace hf CLI: search/download/upload models, datasets.
segment-anything-model	mlops	SAM: zero-shot image segmentation via points, boxes, masks.
weights-and-biases	mlops	W&B: log ML experiments, sweeps, model registry, dashboards.
evaluating-llms-harness	mlops	lm-eval-harness: benchmark LLMs (MMLU, GSM8K, etc.).
llama-cpp	mlops	llama.cpp local GGUF inference + HF Hub model discovery.
obliteratus	mlops	OBLITERATUS: obliterate LLM refusals (diff-in-means).
serving-llms-vllm	mlops	vLLM: high-throughput LLM serving, OpenAI API, quantization.
outlines	mlops	Outlines: structured JSON/regex/Pydantic LLM generation.
strategic-reading	uncategorized	Read a book, article, transcript, or case study through the lens of a specific strategic problem you
gif-search	media	Search/download GIFs from Tenor via curl + jq.
spotify	media	Spotify: play, search, queue, manage playlists and devices.
youtube-content	media	YouTube transcripts to summaries, threads, blogs.
article-enrichment	uncategorized	Transform raw article text dumps in the brain into structured pages with executive summary, verbatim quotes, key insights, why-it-matters, a
llm-wiki	research	Karpathy
pixelrag-mcp-operations	research	Operate and harden a local PixelRAG MCP wrapper for Hermes without exposing dangerous long-lived serve helpers through the gateway.
blogwatcher	research	Monitor blogs and RSS/Atom feeds via blogwatcher-cli tool.
[REDACTED]	research	Evaluate whether a third-party service or product is trustworthy, integrable, and compliant enough for automation, purchasing, or review for

Skill	Category	Public-safe description
[REDACTED]	research	Operate the academic research workflow from paper discovery through experiment-backed writing and submission.
polymarket	research	Query Polymarket: markets, prices, orderbooks, history.
skillpack-check	uncategorized	/
effective-liteparse	uncategorized	Use when a task involves a local PDF, DOCX, PPTX, XLSX, or image and you want the fastest local extraction path. Prefer lit before heavier

Hooks, webhooks, and plugin hook manifests

Hermes has multiple hook-related surfaces: shell-hook allowlists, webhook subscriptions, and imported Claude/Claude plugin hook manifests. The public repo records only surface/count information, not hook command bodies.

Hook manifest surface
.claude/plugins/marketplaces/thedotmack/plugin/hooks/hooks.json
.claude/plugins/marketplaces/thedotmack/cursor-hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/learning-output-style/hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/security-guidance/hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/hookify/hooks/hooks.json
.claude/plugins/marketplaces/claude-plugins-official/plugins/ralph-loop/hooks/hooks.json
[REDACTED]
.claude/plugins/marketplaces/claude-code-warp/plugins/warp/hooks/hooks.json
.claude/plugins/marketplaces/claude-code-plugins/plugins/learning-output-style/hooks/hooks.json
.claude/plugins/marketplaces/claude-code-plugins/plugins/security-guidance/hooks/hooks.json
.claude/plugins/marketplaces/claude-code-plugins/plugins/hookify/hooks/hooks.json
.claude/plugins/marketplaces/claude-code-plugins/plugins/ralph-wiggum/hooks/hooks.json
[REDACTED]
.claude/plugins/marketplaces/ponytail/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.2.0/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.1.0/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.0.1/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.0.0/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.4.0/hooks/hooks.json
.claude/plugins/cache/thedotmack/claude-mem/13.3.0/hooks/hooks.json

Plugin surface

Plugin	Status
browser-browser-use	not enabled
browser-browserbase	not enabled
browser-firecrawl	not enabled
chronos	not enabled
basic	not enabled
nous	not enabled
self-hosted	not enabled
disk-cleanup	not enabled
google_meet	not enabled
fal	not enabled
krea	not enabled
openai	not enabled
openai-codex	not enabled
xai	not enabled
alibaba-provider	not enabled
anthropic-provider	not enabled
arcee-provider	not enabled
bedrock-provider	not enabled
copilot-provider	not enabled
copilot-acp-provider	not enabled
custom-provider	not enabled
deepseek-provider	not enabled
gemini-provider	not enabled
gmi-provider	not enabled
huggingface-provider	not enabled

Plugin	Status
kilocode-provider	not enabled
kimi-coding-provider	not enabled
minimax-provider	not enabled
nous-provider	not enabled
novita-provider	not enabled

Low-Level Surface Files

The repository includes dedicated, low-level public-safe files for each operational surface:

File	Contents
docs/surfaces/tasks.md	Scheduled tasks/cron jobs, modes, scripts, schedules, delivery class, workdir class.
docs/surfaces/skills.md	Skill categories, counts, public-safe examples, operational semantics.
docs/surfaces/hooks.md	Shell hook allowlist state, webhook state, plugin hook manifests.
docs/surfaces/plugins.md	Hermes plugin registry rows and status.
docs/surfaces/mcp-and-toolsets.md	MCP servers and toolset count estimates.
docs/surfaces/models-and-gateway.md	Model routing, gateway status, channel/platform surface.

Model Routing

Role	Provider	Model	Notes
Primary	copilot	gpt-5.4	Default for Telegram/API/CLI gateway sessions
Fallback	copilot	gpt-5.4	Used when primary fails
Optional provider	lmstudio	gemma4unc	http://127.0.0.1:1234/v1
Optional provider	nvidia	meta/llama-3.3-70b-instruct	https://integrate.api.nvidia.com/v1
Optional provider	freetimi	cfbt-kimi	http://127.0.0.1:3271/v1
Optional provider	forge_freetimi	cfbt-kimi	http://127.0.0.1:8081/v1
Optional provider	forge_lmstudio	gemma4unc	http://127.0.0.1:8082/v1

Local model trial status

Item	Status
LM Studio endpoint	not reachable at http://127.0.0.1:1234/v1
Reported model IDs	none
Chat smoke test	blocked_or_unavailable: <urlopen error [Errno 111] Connection refused>
Safety decision	Main Hermes remains openai-codex/gpt-5.5 ; local provider is optional until a model can load reliably.

MCP and External Tooling

MCP server	Transport	Public-safe purpose
gbrain	Local HTTP MCP on 127.0.0.1:3131/mcp	Personal knowledge graph, memory, code graph, facts, schema tools.
notebooklm	npx notebooklm-mcp@latest	Grounded research over registered Google NotebookLM notebooks.
codegraph	Local Node command	Code intelligence over selected repositories.

Scheduled Automation

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Knowledge & memory	4	Keep GBrain/memory/context stores healthy and up to date.
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Current Profiles

The live system currently exposes the default profile publicly as:

Profile	Model	Gateway	Alias	Distribution
◆default	gpt-5.4	running	—	—

Future recommended profile split:

Profile	Purpose	Model stance
default	Main Telegram/personal assistant	openai-codex/gpt-5.5 primary, Copilot fallback.
local-lmstudio	Experimental local-model testing	LM Studio only after a local model can pass smoke tests.
dev	Developer workflows with larger context and coding tools	Can selectively use cloud or local providers.
voice-coach	Low-latency full-duplex voice experiments	Minimal tools/skills, local STT/TTS first.

Reliability and Safety Boundaries

- Public repo does **not** publish secrets, chat IDs, API keys, OAuth tokens, cookies, private finance details, or client-specific payloads.
- The gateway is supervised by systemd and multiple watchdog cron jobs.
- GBrain runs through a shared local HTTP MCP server to avoid PGLite lock contention.
- Backups and sync jobs run through cron with Google Drive/GitHub targets.
- Windows is treated as a remote worker controlled via SSH, not as the primary Hermes control plane.
- Experimental providers are explicitly optional and must pass direct API + Hermes smoke tests before being used for real agent work.

Operational Notes

- Hermes version/status summary:

```
Hermes Agent v0.17.0 (2026.6.19) · upstream d6cf383d
Project: ~/.hermes/hermes-agent
Python: 3.11.15
OpenAI SDK: 2.24.0
Update available: 70 commits behind – run 'hermes update'
```

- Fallback chain:

```
Primary:  gpt-5.4  (via copilot)

Fallback chain (1 entry):
1. gpt-5.4  (via copilot)

Tried in order when the primary fails (rate-limit, 5xx, connection errors).
Docs: https://hermes-agent.nousresearch.com/docs/user-guide/features/fallback-providers
```

- MCP list:

MCP Servers:

Name	Transport	Tools	Status
codegraph	~/.nvm/versions/no... all		enabled
gbrain	http://127.0.0.1:3131/mcp	all	enabled
notebooklm	<code>npx -y notebooklm-mcp@latest</code>	all	enabled
homeway	https://homeway.io/api/mcp	all	enabled
windows-cua	~/.local/bin/windo... all		enabled
monarch	https://api.monarch.com/mcp	17 selected	enabled
pixelrag	~/github/pixelrag-... all		enabled

Maintenance

This repository is regenerated by `scripts/update_architecture.py`. The updater is designed to be safe for a public repository:

1. collect local Hermes runtime state,
2. redact secrets and private identifiers,
3. group sensitive scheduled jobs by category,
4. regenerate Markdown, SVG, HTML, JSON inventory, and PDF,
5. commit and push only if files changed.

Source Files

- `ARCHITECTURE.md` - canonical public architecture document.
- `docs/hermes-architecture.svg` - standalone diagram.
- `docs/hermes-architecture.html` - browser-friendly diagram page.
- `docs/Hermes-Architecture.pdf` - rendered PDF copy.
- `docs/surfaces/*.md` - low-level public-safe inventories for tasks, skills, hooks, plugins, MCP/toolsets, and gateway/model routing.
- `data/inventory.public.json` - redacted machine-readable snapshot.
- `scripts/update_architecture.py` - public-safe regeneration script.

Rendered from GBrain markdown via Hermes brain-pdf.